(i)

$$F_{k+1} = F_k + F_{k-1} \leq 2F_k$$

$$F_{n+2} \leq 2F_{n+1} \leq 2^2 F_n$$

$$F_r \leq 2^{r-n} F_n \qquad (r \geq n)$$

(ii)

$$\begin{aligned}\sum_{r=1}^{n} \frac{F_{r+1}}{10^{r-1}} &= \sum_{r=2}^{n+1} \frac{F_r}{10^{r-2}} \\ &= 100 \sum_{r=2}^{n+1} \frac{F_r}{10^r} \\ &= 100S_n - 10F_1 + \frac{F_{n+1}}{10^{n-1}}\end{aligned}$$

$$\sum_{r=1}^{n} \frac{F_r}{10^{r-1}} = 10S_n$$

$$\sum_{r=1}^{n} \frac{F_{r-1}}{10^{r-1}} = \sum_{r=0}^{n-1} \frac{F_r}{10^r} = F_0 + S_n - \frac{F_n}{10^n}$$

$$\begin{aligned}&\sum_{r=1}^{n} \frac{F_{r+1}}{10^{r-1}} - \sum_{r=1}^{n} \frac{F_r}{10^{r-1}} - \sum_{r=1}^{n} \frac{F_{r-1}}{10^{r-1}} \\ &= 100S_n - 10F_1 + \frac{F_{n+1}}{10^{n-1}} - 10S_n - \left(F_0 + S_n - \frac{F_n}{10^n}\right) \\ &= 89S_n - 10F_1 - F_0 + \frac{F_n}{10^n} + \frac{F_{n+1}}{10^{n-1}}\end{aligned}$$

(iii)

$$\because F_{r+1} - F_r - F_{r-1} = 0$$

$$\therefore 0 = 89S_n - 10F_1 - F_0 + \frac{F_n}{10^n} + \frac{F_{n+1}}{10^{n-1}}$$

$$F_1 = 1,\ F_0 = 0$$

$$89S_n = 10 - \frac{F_n}{10^n} - \frac{F_{n+1}}{10^{n-1}}$$

$$\because F_n \leq 2^{n-1}$$

$$\therefore \frac{F_n}{10^n} \leq \frac{2^{n-1}}{10^n} = \frac{1}{2}\left(\frac{1}{5}\right)^n \to 0$$

Similarly, $\frac{F_{n+1}}{10^{n-1}} \to 0$

$$\therefore \sum_{r=1}^{\infty} \frac{F_r}{10^r} = \frac{10}{89}$$

$$F_7 = 13$$

For $r \geq 7$,

$$F_r \leq 2^{r-7} F_7 = 13 \cdot 2^{r-7}$$

$$\begin{aligned}
\sum_{r=7}^{\infty} \frac{F_r}{10^r} &\leq \sum_{r=7}^{\infty} \frac{13 \cdot 2^{r-7}}{10^r} \\
&= \frac{13}{10^7}\left(1 + \frac{1}{5} + \frac{1}{5^2} + \cdots\right) \\
&= \frac{13}{10^7} \cdot \frac{5}{4} \\
&< 2 \times 10^{-6}
\end{aligned}$$

$$S_6 = \frac{1}{10} + \frac{1}{10^2} + \frac{2}{10^3} + \frac{3}{10^4} + \frac{5}{10^5} + \frac{8}{10^6} = 0.112358$$

$$\because S_6 < \frac{10}{89} = S_6 + \sum_{r=7}^{\infty} \frac{F_r}{10^r} < 0.112358 + 10^{-6}$$

$$0.112358 < \frac{10}{89} < 0.112360 \implies 0.0112358 < \frac{1}{89} < 0.0112360$$

(iv) Let $T_n = \sum_{r=1}^{n} \frac{F_r}{100^r}$

$$0 = \sum_{r=1}^{n} \frac{F_{r+1} - F_r - F_{r-1}}{100^{r-1}}$$

$$\sum_{r=1}^{n} \frac{F_{r+1}}{100^{r-1}} = 10000T_n - 100F_1 + \frac{F_{n+1}}{100^{n-1}}$$

$$\sum_{r=1}^{n} \frac{F_r}{100^{r-1}} = 100T_n$$

$$\sum_{r=1}^{n} \frac{F_{r-1}}{100^{r-1}} = F_0 + T_n - \frac{F_n}{100^n}$$

$$0 = 9899T_n - 100 + \frac{F_n}{100^n} + \frac{F_{n+1}}{100^{n-1}}$$

$$\implies \frac{1}{9899} = \sum_{r=1}^{\infty} \frac{F_r}{100^{r+1}}$$

$$\begin{aligned}\frac{1}{9899} = \frac{1}{100^2} + \frac{1}{100^3} + \frac{2}{100^4} + \frac{3}{100^5} + \frac{5}{100^6} + \frac{8}{100^7} \\ + \frac{13}{100^8} + \frac{21}{100^9} + \frac{34}{100^{10}} + \frac{55}{100^{11}} + \sum_{k=11}^{\infty} \frac{F_k}{100^{k+1}}\end{aligned}$$

$$\begin{aligned}\sum_{k=11}^{\infty} \frac{F_k}{100^{k+1}} &\leq \sum_{r=11}^{\infty} \frac{89 \cdot 2^{r-11}}{100^{r+1}} \\ &= \frac{89}{100^{12}} \left(1 + \frac{1}{50} + \frac{1}{50^2} + \cdots\right) \\ &< \frac{1}{100^{11}}\end{aligned}$$

$$\therefore \frac{1}{9899} = 0.000101020305081321345 5\ldots$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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